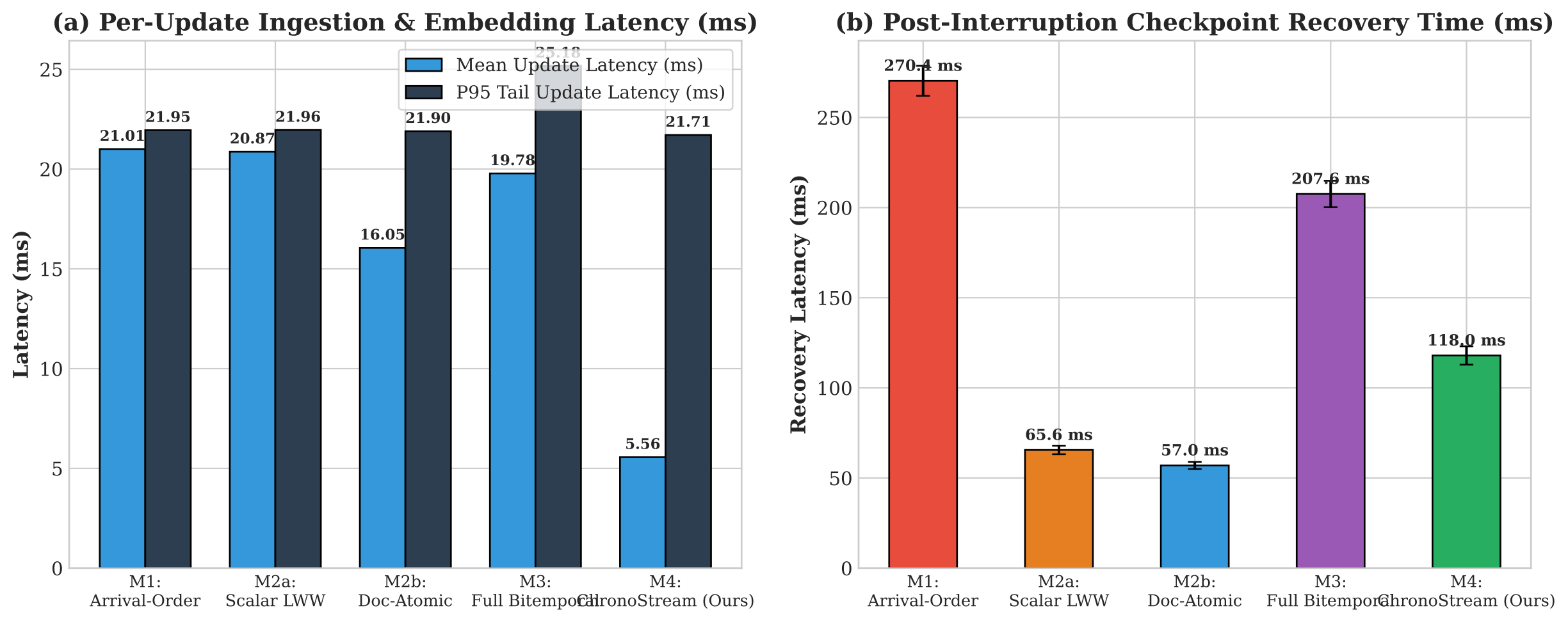
(a) Per-Update Ingestion & Embedding Latency (ms)
Mean Update Latency (ms)
P95 Tail Update Latency (ms)
Latency (ms)
0
5
10
15
20
25
21.01
21.95
20.87
21.96
16.05
21.90
19.78
25.18
5.56
21.71
M1:
Arrival-Order
M2a:
Scalar LWW
M2b:
Doc-Atomic
M3:
Full Bitempor
M4:
ChronoStream (Ours)
(b) Post-Interruption Checkpoint Recovery Time (ms)
Recovery Latency (ms)
0
50
100
150
200
250
270.4 ms
65.6 ms
57.0 ms
207.6 ms
118.0 ms
M1:
Arrival-Order
M2a:
Scalar LWW
M2b:
Doc-Atomic
M3:
Full Bitempor
M4:
ChronoStream (Ours)